

Question	Answer	Marks
9(a)(i)	<i>either:</i> cannot predict when a (particular) nucleus will decay <i>or:</i> cannot predict which nucleus will decay next	B1
9(a)(ii)	not affected by external / environmental factors	B1
9(b)	time for activity to halve	B1
9(c)	energy = $(189 \times 7.826) + (4 \times 7.074) - (193 \times 7.774)$	C1
	= 7.03eV	A1
9(d)(i)	decay constant	A1
9(d)(ii)	decay constant / magnitude of gradient = $1.4 / 0.84$	C1
	half-life = $\ln 2 / (1.4 / 0.84)$ = 0.42ms	A1
9(e)(i)	positrons collide with electrons and annihilate	B1
9(e)(ii)	long enough to have time to conduct investigation, not so long as to cause patient <u>unnecessary</u> exposure to radiation	B1

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